

Backup and Recovery Procedures-67

1. Introduction

Backup and recovery procedures define how data is protected and restored when loss or corruption occurs. A backup ensures data availability in case of hardware failure, cyberattack, or accidental deletion. Recovery ensures that this backed-up data is restored correctly within the required timeframes. Together, they are the backbone of any business continuity plan.

2. Objectives of Backup and Recovery Procedures

- Ensure data integrity and availability.
- Minimize downtime during failures.
- Meet RPO (Recovery Point Objective) and RTO (Recovery Time Objective).
- Comply with legal and industry regulations.
- Provide clear, tested steps for IT staff to follow.

3. Components of Backup Procedures

1. Backup Planning

- Identify critical systems and data.
- Define backup types: full, incremental, differential, or continuous.
- Set schedules: hourly, daily, weekly, monthly.
- Align with RPO and RTO.

2. Backup Execution

- Automate backups with tools (e.g., Veeam, Commvault, Acronis, native OS utilities).
- Store copies on multiple media (disk, tape, cloud, hybrid).
- Use encryption to secure backup data.

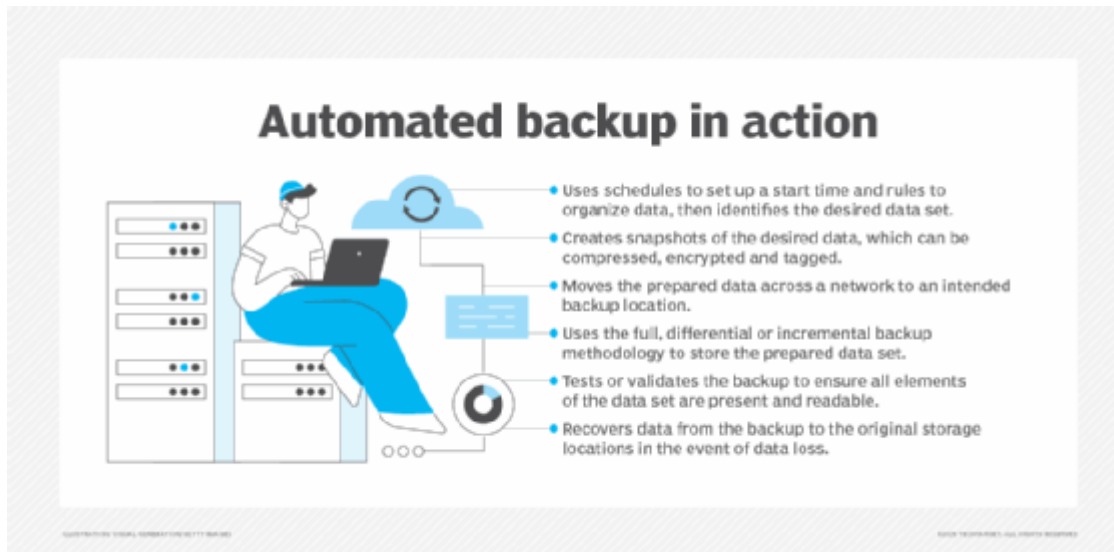
3. Backup Monitoring

- Generate logs and reports for each backup run.

- Alert administrators on failures.
- Monitor storage capacity to prevent incomplete jobs.

4. Backup Testing

- Perform test restores to confirm data is recoverable.
- Validate integrity of backup files.
- Document test results for audits.



4. Components of Recovery Procedures

1. Recovery Planning

- Define priority order for restoring systems.
- Create recovery runbooks (step-by-step restoration guides).
- Assign responsibilities to recovery teams.

2. Recovery Steps

- Identify the type of incident: hardware failure, ransomware, user error.
- Select appropriate recovery point based on RPO.
- Restore data from local, offsite, or cloud storage.
- Validate recovered data for completeness.

3. Recovery Testing

- Simulate disaster scenarios to measure RTO.
- Test both full system restores and granular restores (single files, databases).

- Refine procedures based on test outcomes.

4. Post-Recovery Review

- Document incident details.
- Analyze gaps in backup or recovery steps.
- Update procedures to prevent future failures.

5. Example Backup and Recovery Workflow

1. Backup Procedure

- Full backup every Sunday.
- Incremental backups daily at midnight.
- Copies stored onsite on NAS and offsite in AWS cloud.
- Data encrypted with AES-256.

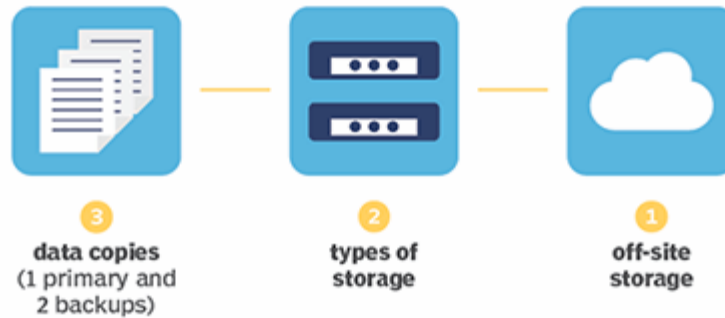
2. Recovery Procedure (Database Crash Example)

- Detect failure and notify admin.
- Stop affected application services.
- Restore latest full backup from Sunday.
- Apply incremental backups from Monday to failure time.
- Validate database consistency.
- Restart services and confirm application functionality.

6. Best Practices for Backup and Recovery

- Follow the **3-2-1 rule**: 3 copies of data, 2 different media, 1 offsite.

3-2-1 backup strategy steps



- Use immutable backups to protect against ransomware.
- Automate both backup and recovery testing.
- Document and train staff on procedures.
- Review procedures quarterly or after major system changes.

Backup and recovery procedures ensure business continuity by protecting critical data and providing a structured way to restore it after failure. Backup procedures focus on capturing and storing data, while recovery procedures focus on restoring it within acceptable timeframes. Effective policies, regular testing, and documentation guarantee that an organization can withstand data loss and quickly resume operations.